

Администрирование сетевых подсистем

Лабораторная работа №7 — Настройка firewalld, Port Forwarding и Masquerading в Linux

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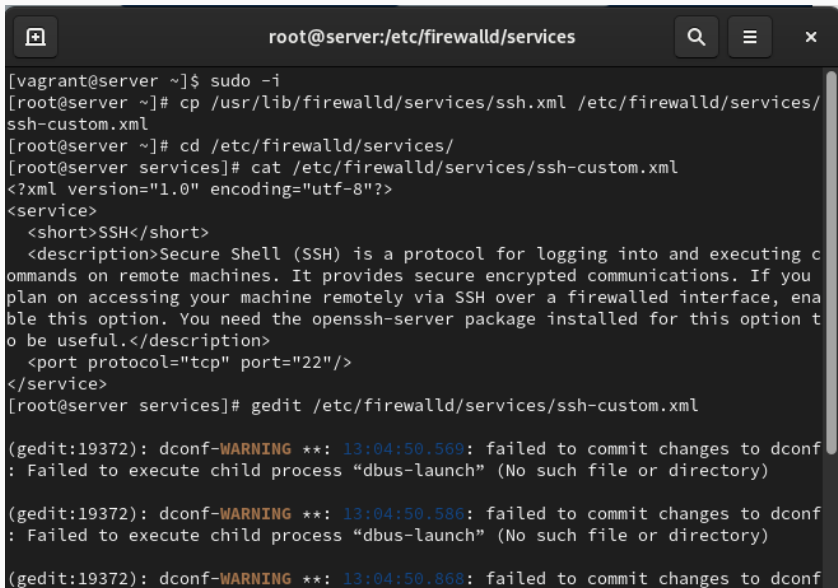
Цели и задачи работы

Цель лабораторной работы

Получить практические навыки настройки межсетевого экрана `firewalld` в части создания служб, перенаправления портов и активации Masquerading.

Выполнение работы

Копирование и просмотр стандартной службы SSH



```
root@server:/etc/firewalld/services

[vagrant@server ~]$ sudo -i
[root@server ~]# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/services/ssh-custom.xml
[root@server ~]# cd /etc/firewalld/services/
[root@server services]# cat /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing c
ommands on remote machines. It provides secure encrypted communications. If you
plan on accessing your machine remotely via SSH over a firewalled interface, ena
ble this option. You need the openssh-server package installed for this option t
o be useful.</description>
  <port protocol="tcp" port="22"/>
</service>
[root@server services]# gedit /etc/firewalld/services/ssh-custom.xml

(gedit:19372): dconf-WARNING **: 13:04:50.569: failed to commit changes to dconf
: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:19372): dconf-WARNING **: 13:04:50.586: failed to commit changes to dconf
: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:19372): dconf-WARNING **: 13:04:50.868: failed to commit changes to dconf
```

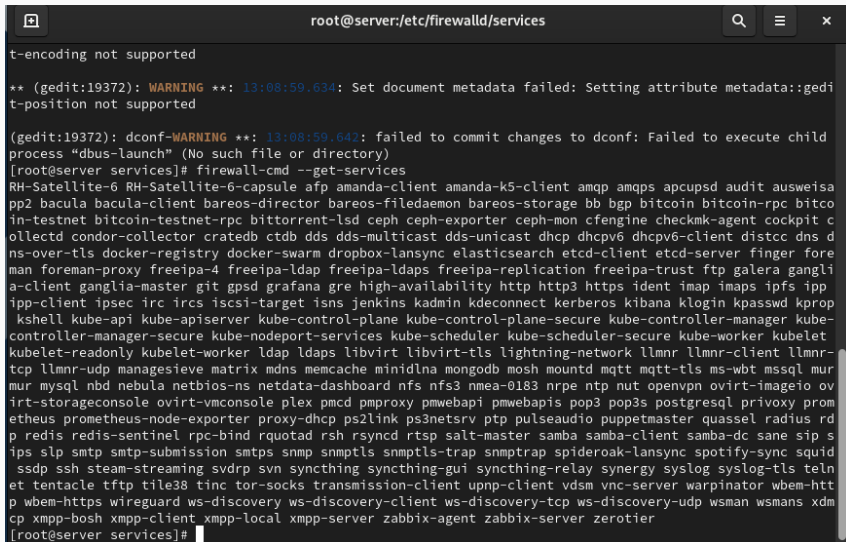
Внесение изменений в файл службы



```
1 <?xml version="1.0" encoding="utf-8"?>
2 <service>
3   <short>SSH</short>
4   <description>This a modified file</description>
5   <port protocol="tcp" port="2022"/>
6 </service>
```

Рис. 2: Редактирование службы ssh-custom.xml

Проверка списка служб и добавление новой

A terminal window titled 'root@server:/etc/firewalld/services' with standard window controls (minimize, maximize, close) and a search icon. The terminal output shows several warning messages from gedit and a long list of services currently configured in the firewall. The list includes services like RH-Satellite-6, amanda-client, amqp, amqps, apcupsd, audit, ausweis, pp2, bacula, bareos, bb, bgp, bitcoin, bitcoind, bitcoind-rpc, bitcoind-testnet, bittorrent, ceph, ceph-exporter, ceph-mon, cfengine, checkmk-agent, cockpit, collectd, condor, cratedb, ctdb, dds, dds-multicast, dds-unicast, dhcp, dhcpv6, dhcpv6-client, distcc, dns, dns-over-tls, docker, docker-registry, docker-swarm, dropbox, dropbox-lansync, elasticsearch, etcd-client, etcd-server, finger, foreman, freeipa, freeipa-ldap, freeipa-ldaps, freeipa-replication, freeipa-trust, ftp, galera, ganglia, git, gpsd, grafana, gre, high-availability, http, http3, https, ident, imap, imaps, ipfs, ipp, ipp-client, ipsec, irc, ircs, iscsi-target, isns, jenkins, kadmin, kdeconnect, kerberos, kibana, klogin, kpasswd, kprop, kshell, kube-api, kube-apiserver, kube-control-plane, kube-control-plane-secure, kube-controller-manager, kube-controller-manager-secure, kube-nodeport-services, kube-scheduler, kube-scheduler-secure, kube-worker, kubelet, kubelet-readonly, kubelet-worker, ldap, ldaps, libvirt, libvirt-tls, lightning-network, llmnr, llmnr-client, llmnr-tcp, llmnr-udp, managesieve, matrix, mdns, memcached, minidlna, mongodb, mosh, mountd, mqtt, mqtt-tls, ms-wbt, mssql, murmur, mysql, nbd, nebula, netbios-ns, netdata-dashboard, nfs, nfs3, nmap, nmap-0183, nmap, ntp, nut, openvpn, ovirt-imageio, ovirt-storageconsole, ovirt-vmconsole, plex, pmcd, pmproxy, pmwebapi, pmwebapis, pop3, pop3s, postgresql, privoxy, prometheus, prometheus-node-exporter, proxy-dhcp, ps2link, ps3netsrv, ptp, pulseaudio, puppetmaster, quassel, radius, redis, redis-sentinel, rpc-bind, rquotad, rsh, rsyncd, rtsp, salt-master, samba, samba-client, samba-dc, sane, sip, sipps, slp, smtp, smtp-submission, smtps, snmp, snmptls, snmptls-trap, snmptrap, spideroak-lansync, spotify-sync, squid, sssd, ssh, steam-streaming, svdrp, svn, syncthing, syncthing-gui, syncthing-relay, synergy, syslog, syslog-tls, telnet, tftpd, tftp, tinc, tor-socks, transmission-client, unpn-client, vdsm, vnc-server, warpinator, wbm-http, wbm-https, wireguard, ws-discovery, ws-discovery-client, ws-discovery-tcp, ws-discovery-udp, wsman, wsmans, xdmcp, xmpp-bosh, xmpp-client, xmpp-local, xmpp-server, zabbix-agent, zabbix-server, and zerotier.

```
root@server:/etc/firewalld/services
t-encoding not supported
** (gedit:19372): WARNING **: 13:08:59.634: Set document metadata failed: Setting attribute metadata::gedit-position not supported
(gedit:19372): dconf-WARNING **: 13:08:59.642: failed to commit changes to dconf: Failed to execute child process "dbus-launch" (No such file or directory)
[root@server services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweis
pp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoind bitcoind-rpc bitcoind-testnet bittorrent ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit c
ollectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns d
ns-over-tls docker-docker-registry docker-swarm dropbox dropbox-lansync elasticsearch etcd-client etcd-server finger fore
man foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp
ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop
kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-
controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet
kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-
tcp llmnr-udp managesieve matrix mdns memcached minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql mur
mur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmap-0183 nmap ntp nut openvpn ovirt-imageio ov
irt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prom
etheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rd
p redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip s
ips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid
ssdp ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls teln
et tftpd tftp tinc tor-socks transmission-client unpn-client vdsm vnc-server warpinator wbm-http wbm-htt
p wbm-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdm
cp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]#
```

Port Forwarding 2022 → 22

```
root@server:/etc/firewalld/services

cp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --reload
bash: firewall-cmd: command not found...
[root@server services]# firewall-cmd --reload
success
[root@server services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweis
op2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitco
in-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit c
ollectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns d
ns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger fore
man foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp
ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop
kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-
controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet
kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-
tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql mur
mur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmap-0183 nrpe ntp nut openvpn ovirt-imageio ov
irt-storageconsole ovirt-vmconsole plex pmdc pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prom
etheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rd
p redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip s
ips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid
ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog sysl
og-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client unpn-client vdsms vnc-server warpinat
or wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman
wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh
[root@server services]#
```


Port Forwarding 2022 → 22

```
root@server:/etc/firewalld/services

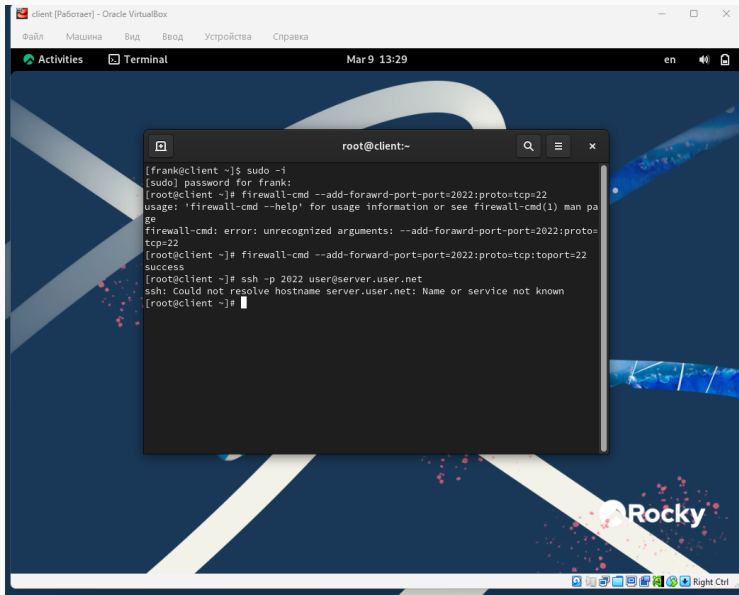
cp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --reload
bash: firewall-cmd: command not found...
[root@server services]# firewall-cmd --reload
success
[root@server services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweis
op2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitco
in-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit c
ollectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns d
ns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger fore
man foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp
ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop
kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-
controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet
kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-
tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql mur
mur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ov
irt-storageconsole ovirt-vmconsole plex pncd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prom
etheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rd
p redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip s
ips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid
ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog sysl
og-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client unpn-client vdsms vnc-server warpinat
or wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman
wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh
[root@server services]#
```

Проверка параметров ядра

```
wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh
[root@server services]# firewall-cmd --add-service=ssh-custom
success
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server services]# firewall-cmd --add-service=ssh-custom --permanent
success
[root@server services]# firewall-cmd --reload
success
[root@server services]#
```

Рис. 6: Проверка sysctl параметра ip_forward

Включение forwarding и masquerading



The screenshot shows a terminal window within an Oracle VM VirtualBox environment. The host is named 'client [Работаer]'. The terminal is running on a Rocky Linux system. The user 'frank' is logged in and uses 'sudo' to become root. The root user then attempts to configure the firewall using 'firewall-cmd'. The first command, 'firewall-cmd --add-forward-port-port=2022:proto=tcp=22', results in an error: 'error: unrecognized arguments: --add-forward-port-port=2022:proto=tcp=22'. The second command, 'firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22', is successful. Finally, the user attempts to connect to 'server.user.net' via SSH on port 2022, but the connection fails with the message 'ssh: Could not resolve hostname server.user.net: Name or service not known'.

```
client [Работаer] - Oracle VM VirtualBox
Файл  Машина  Вид  Ввод  Устройства  Справка
Activities  Terminal  Mar 9 13:29  en  [Speaker Icon]  [Document Icon]

root@client:~
[frank@client ~]$ sudo -i
[sudo] password for frank:
[root@client ~]# firewall-cmd --add-forward-port-port=2022:proto=tcp=22
usage: 'firewall-cmd --help' for usage information or see firewall-cmd(1) man page
firewall-cmd: error: unrecognized arguments: --add-forward-port-port=2022:proto=tcp=22
[root@client ~]# firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22
success
[root@client ~]# ssh -p 2022 user@server.user.net
ssh: Could not resolve hostname server.user.net: Name or service not known
[root@client ~]#
```

Проверка доступа в Интернет с клиента

```
root@server:/etc/firewalld/services

[root@server services]# firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22
success
[root@server services]# ssh -p 2022 user@server.user.net
ssh: connect to host server.user.net port 2022: Connection refused
[root@server services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 0
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
```

Создание provisioning-скрипта firewall.sh

```
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
net.ipv6.conf.lo.forwarding = 0
net.ipv6.conf.lo.mc_forwarding = 0
[root@server services]# echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
[root@server services]# sysctl -p /etc/sysctl.d/90-forward.conf
net.ipv4.ip_forward = 1
[root@server services]# firewall-cmd --zone=public --add-masquerade --permanent
success
[root@server services]# firewall-cmd --reload
success
[root@server services]#
```

Рис. 9: Скрипт firewall.sh

Выводы по работе

- Создана пользовательская служба firewalld на базе SSH.
- Настроено перенаправление порта 2022 → 22.
- Включены IPv4 forwarding и masquerading.
- Проверен доступ клиента во внешнюю сеть.
- Подготовлены provisioning-файлы для автоматизации конфигурации Firewalld.